Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software engineer with 2+ years building cloud applications, secure APIs, cross-platform workflows, and production integrations. Experienced in Python, JavaScript, TypeScript, AWS, Azure, cloud storage, SQL, automated testing, CI/CD, webhooks, and reliable SaaS delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python cloud services for 500+ users using Azure SQL and Blob Storage, tuning indexed queries and secure access paths to reduce application response latency from 1.2 seconds to 350 milliseconds.
- Implemented asynchronous workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving reliability, observability, maintainability, and production issue investigation support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, integrating secure uploads, authentication, job-state updates, and grading workflows while investigating production failures affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress reporting, and failure handling, reducing turnaround 60% while maintaining reliable application behavior under concurrent production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed cross-platform Python services and REST integrations using AWS Glue, S3, SQL, filtering, and pagination, improving response times 30% and data-processing efficiency 68% across production cloud environments.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating application audits, defect diagnosis, release validation, and reliable deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight services with authentication, WebSocket messaging, and state synchronization, coordinating 2 million+ Azure SQL records through secure interfaces and real-time updates.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving software quality, deployment consistency, and operational visibility.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built cross-platform booking workflows and Flask APIs for reservations, scheduling, status updates, and payments across 50,000+ records, translating requirements into maintainable application features with secure Firebase access.
- Implemented Firebase persistence, webhooks, and payment synchronization with structured validation and logging, reducing manual payment handling 70% while improving integration reliability, issue visibility, and responsive released-application support.

Skills

Languages & Applications: Python, Java, JavaScript, TypeScript, SQL, React, Next.js, Node.js, Flask, FastAPI, responsive interfaces, cross-platform development
Cloud & Integration: AWS, AWS RDS, AWS Glue, Amazon S3, Azure, Blob Storage, REST APIs, WebSockets, webhooks, API integration, asynchronous workflows, distributed workflows
Data: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, relational databases, data modeling, joins, filtering, pagination, query optimization
Quality & Delivery: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, automated testing, unit testing, integration testing, application testing, debugging, logging, monitoring, code review, production support, Agile

Projects

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built secure payment workflows supporting 50,000+ users and one million transactions, integrating REST APIs, WebSocket updates, MongoDB persistence, validation, monitoring, and automated Kubernetes deployment through reliable CI/CD pipelines.

Chat2SQL | Code | *Java, TypeScript, PostgreSQL, MySQL, MongoDB, Docker*
- Built a Dockerized cloud-ready application supporting natural-language queries across PostgreSQL, MySQL, and MongoDB, implementing REST APIs, input validation, relational joins, and reproducible Maven builds for reliable deployment.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built a responsive Next.js and TypeScript application using reusable components, secure authentication, MongoDB persistence, Stripe integration, and AI-assisted image workflows across desktop and mobile device form factors.